```
clc;
clear;
close all;
% Cell Centers (3 Cells)
cellPos = [0 500 1000];
% User Movement
user = 0:10:1000;
% Nearest Cell Selection (Handover)
handover = zeros(size(user));
for i = 1:length(user)
    [~,handover(i)] = min(abs(user(i)-cellPos));
end
% Display
disp('User Position Connected Cell')
```

```
User Position Connected Cell
```

```
disp([user' handover'])
```

```
           0           1
          10           1
          20           1
          30           1
          40           1
          50           1
          60           1
          70           1
          80           1
          90           1
         100           1
         110           1
         120           1
         130           1
         140           1
         150           1
         160           1
         170           1
         180           1
         190           1
         200           1
         210           1
         220           1
         230           1
         240           1
         250           1
         260           2
         270           2
         280           2
         290           2
         300           2
         310           2
         320           2
         330           2
         340           2
         350           2
         360           2
         370           2
         380           2
         390           2
```

```
         400           2
         410           2
         420           2
         430           2
         440           2
         450           2
         460           2
         470           2
         480           2
         490           2
         500           2
         510           2
         520           2
         530           2
         540           2
         550           2
         560           2
         570           2
         580           2
         590           2
         600           2
         610           2
         620           2
         630           2
         640           2
         650           2
         660           2
         670           2
         680           2
         690           2
         700           2
         710           2
         720           2
         730           2
         740           2
         750           2
         760           3
         770           3
         780           3
         790           3
         800           3
         810           3
         820           3
         830           3
         840           3
         850           3
         860           3
         870           3
         880           3
         890           3
         900           3
         910           3
         920           3
         930           3
         940           3
         950           3
         960           3
         970           3
         980           3
         990           3
        1000           3
```

```
% Plot
figure;
```

```
subplot(2,1,1)
plot(user,handover,'b','LineWidth',2)
title('Handover Performance')
xlabel('User Position (m)')
ylabel('Cell Number')
grid on
subplot(2,1,2)


stem(cellPos,[1 2 3],'filled')
hold on
plot(user,ones(size(user))*2,'r--')
title('Cellular Frequency Planning')
xlabel('Distance (m)')
ylabel('Cell')
grid on
```

